



SB GAMES

XXV SIMPÓSIO BRASILEIRO
DE JOGOS E ENTRETENIMENTO
DIGITAL 2026

GOIÂNIA - GO 📍 2026

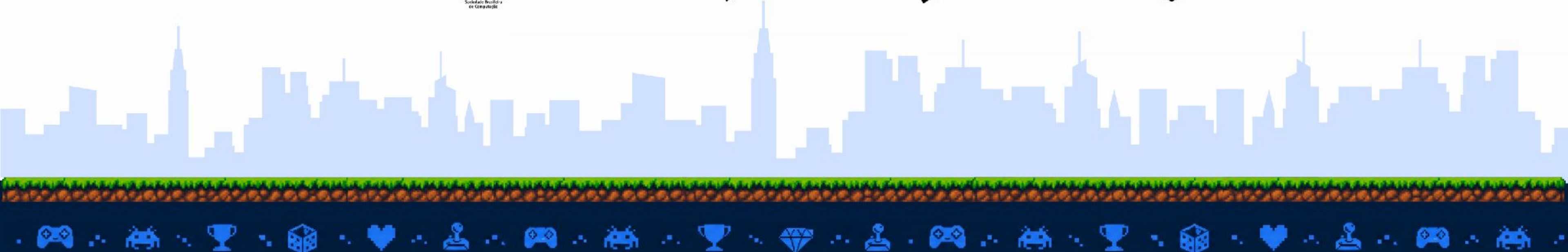
REALIZAÇÃO:



ORGANIZAÇÃO:



INF
INSTITUTO DE
INFORMÁTICA





Computer Vision for MOBA Analytics: A Dataset and Baseline for Visibility Analysis in Dota 2

Ricardo R. Carvalho, Eloísa Oliveira, Luiz B. M. Kummer,

Emerson C. Paraiso, Rayson Laroca

Pontifical Catholic University of Paraná (PUCPR), Curitiba, Brazil

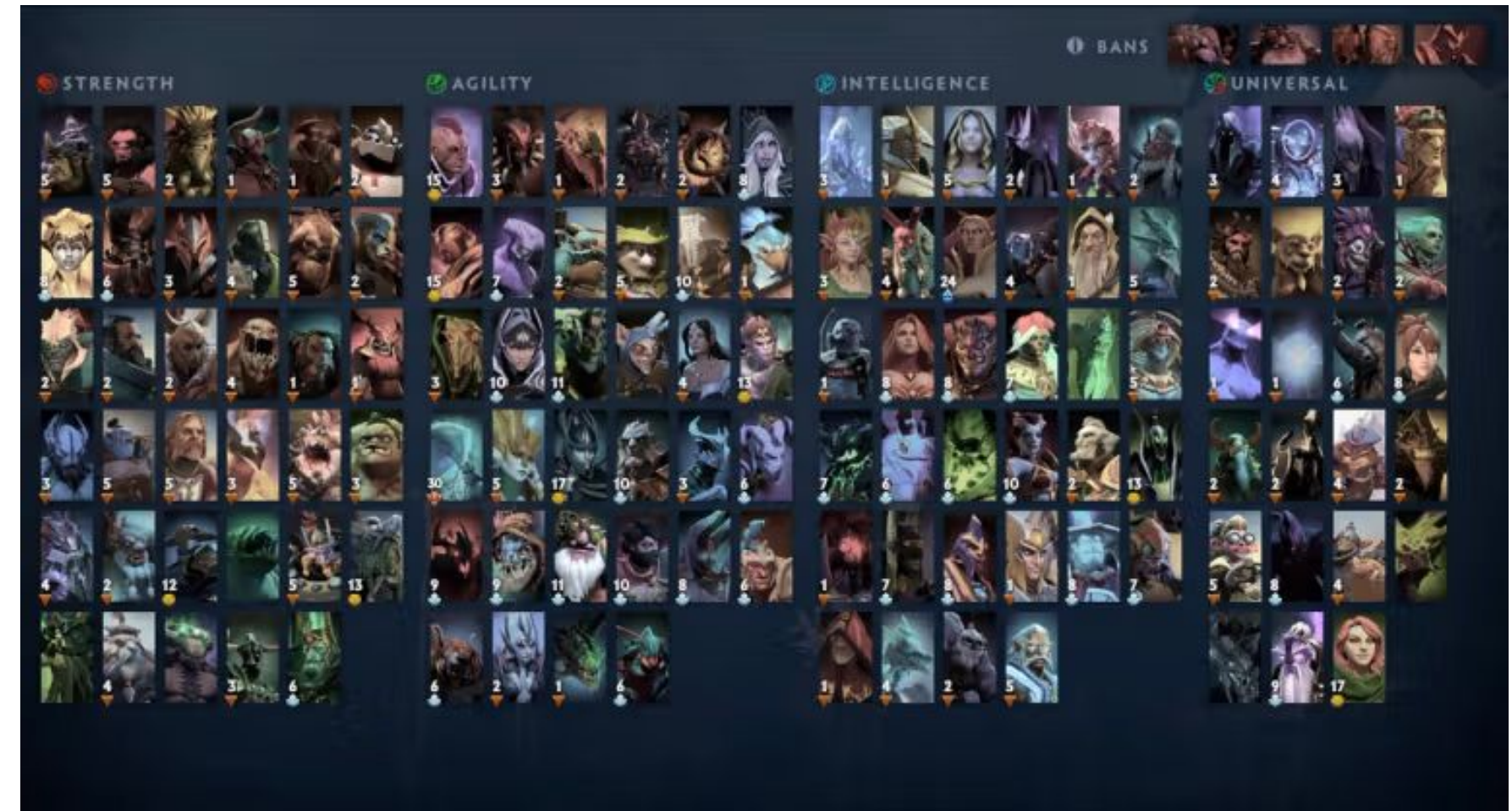


PUCPR
GRUPO MARISTA



Introduction

- Computer vision less explored than structured-data approaches in MOBAs [1]
- MOBAs are especially relevant for artificial intelligence research [2]
- Dota 2 → difficulty to access data from professional LoL matches
- Visibility is a core mechanic of MOBAs as a game of imperfect information



The image displays the heroes available for selection in the game as of patch 7.41e, totaling 126 different characters divided into Strength, Agility, Intelligence, and Universal categories, demonstrating the wide variety of in-game team compositions.

Challenges of Retrieving Visibility from API

Player Vision



.....

Challenges of Retrieving Visibility from API

Player Vision



Real Game State



Challenges of Retrieving Visibility from API

Player Vision



Real Game State





The Dota2-Vis Dataset: Data Collection and Player Detection



Videos from Professional Matches

- Full matches recorded from both teams perspectives
- All matches from the main tournament of the year *The International 2025*
- 144 matches totalizing 288 manually recorded videos
- Future addition of all matches from *The International 2026*, recording ongoing



The image illustrates how the videos are recorded—from the perspective of a single team and following a specific player. The minimap in the bottom-right corner shows that in-game character movements are replicated there, while adhering to the game's visibility rules

Minimaps for Model Training



- **2,477 manually annotated images** retrieved from professional matches
- **21 different classes**, 10 for the players, 10 for clones and one called “others”
 - Approximately **15,000 annotations**
- Challenging and time-consuming process due to **heavy occlusion**

Results Obtained

Results obtained by the different YOLOv11 variants

Model	Precision	Recall	F-Score	mAP50-95
YOLO11n (nano)	0.964	0.881	0.921	0.661
YOLO11s (small)	0.965	0.896	0.929	0.713
YOLO11m (middle)	0.963	0.903	0.932	0.721
YOLO11l (large)	0.974	0.906	0.939	0.729
YOLO11x (extra large)	0.965	0.911	0.937	0.720

*YOLO has become one of the most widely adopted families because it offers a favorable trade-off between inference speed and detection accuracy.

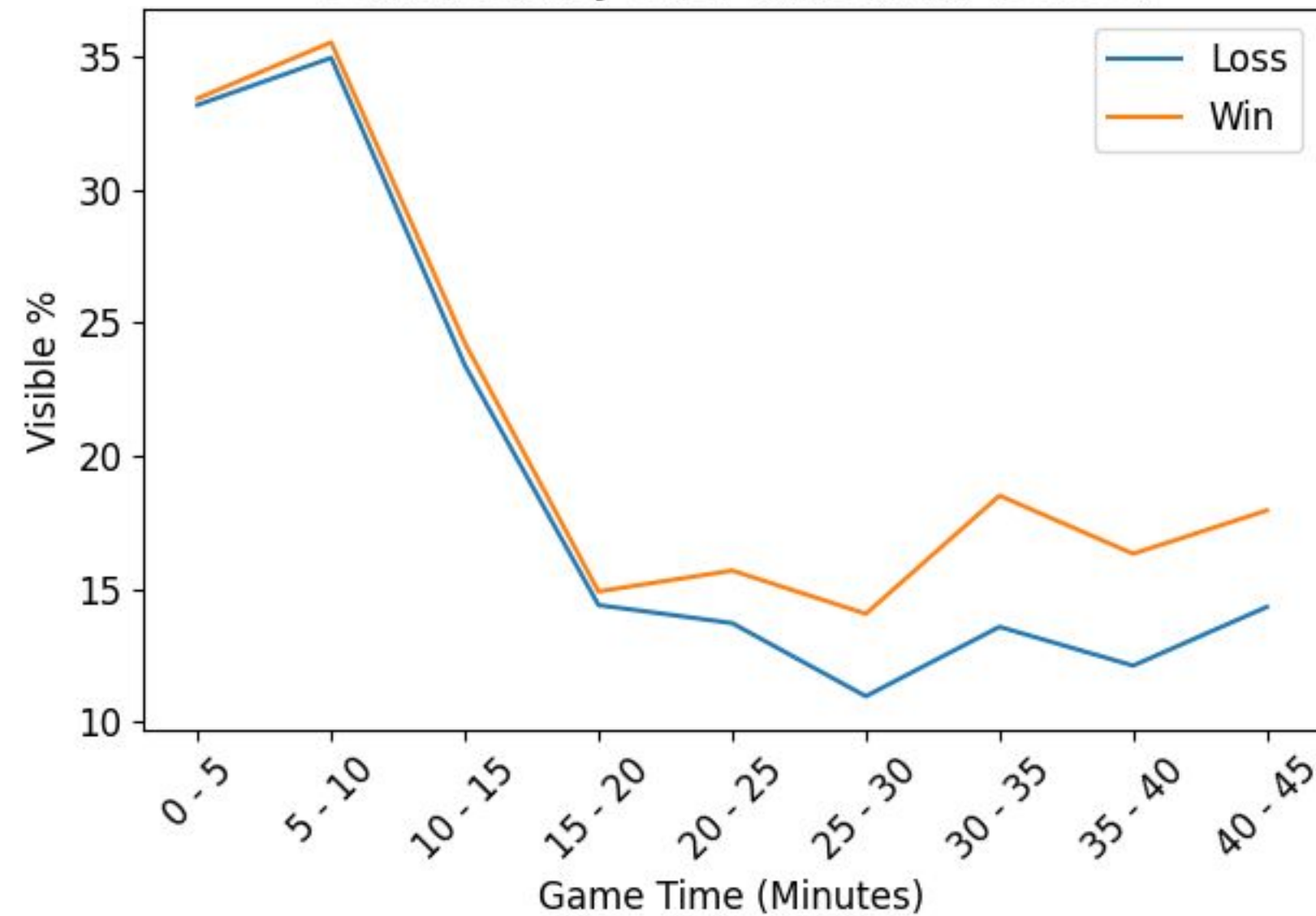
Representative detections produced by YOLO11



Initial Analyses

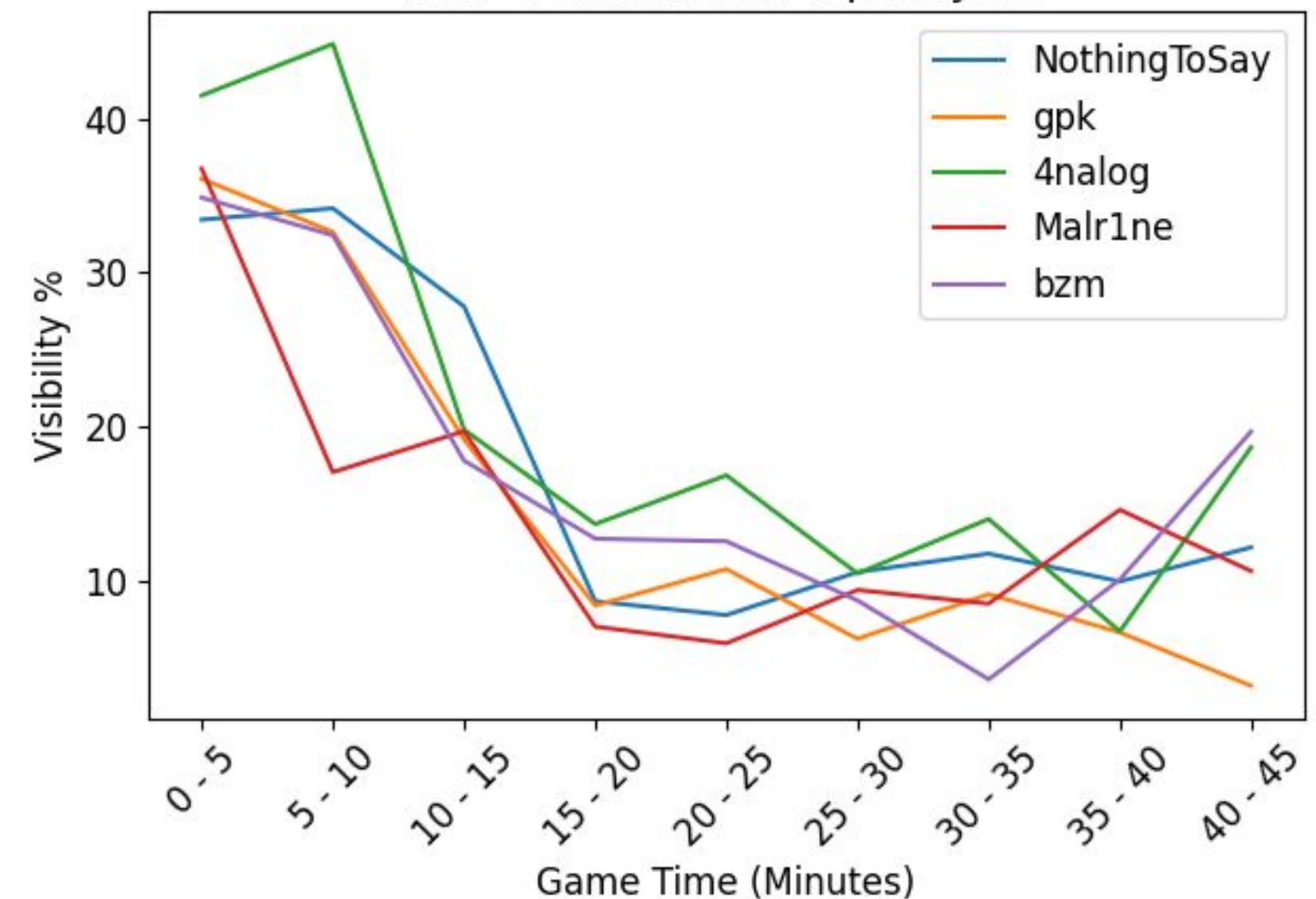
Winning Team x Losing Team

Team Visibility Over Time (Win vs Loss)



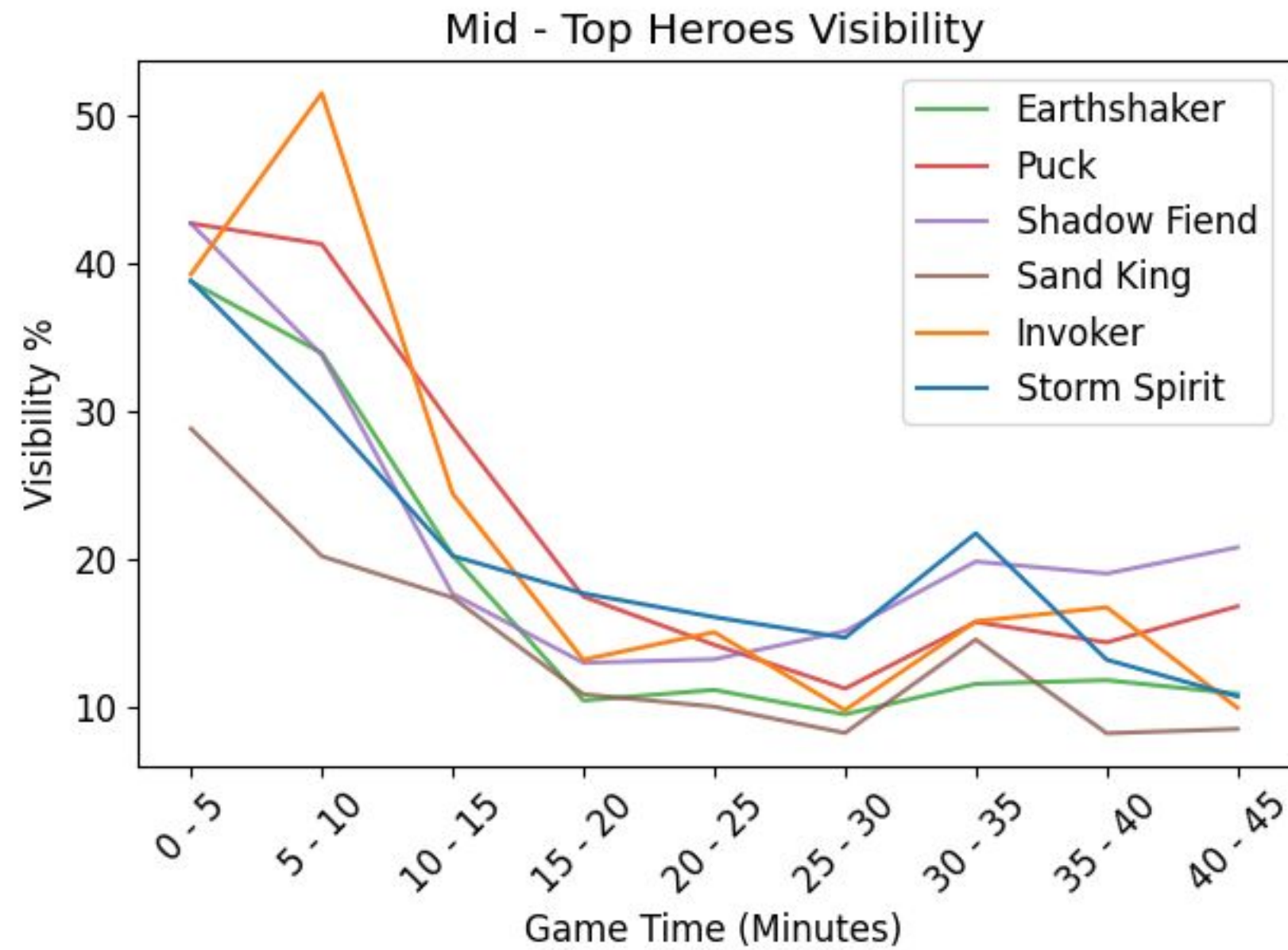
Mid Players as Earthshaker

Mid - Earthshaker Top Players



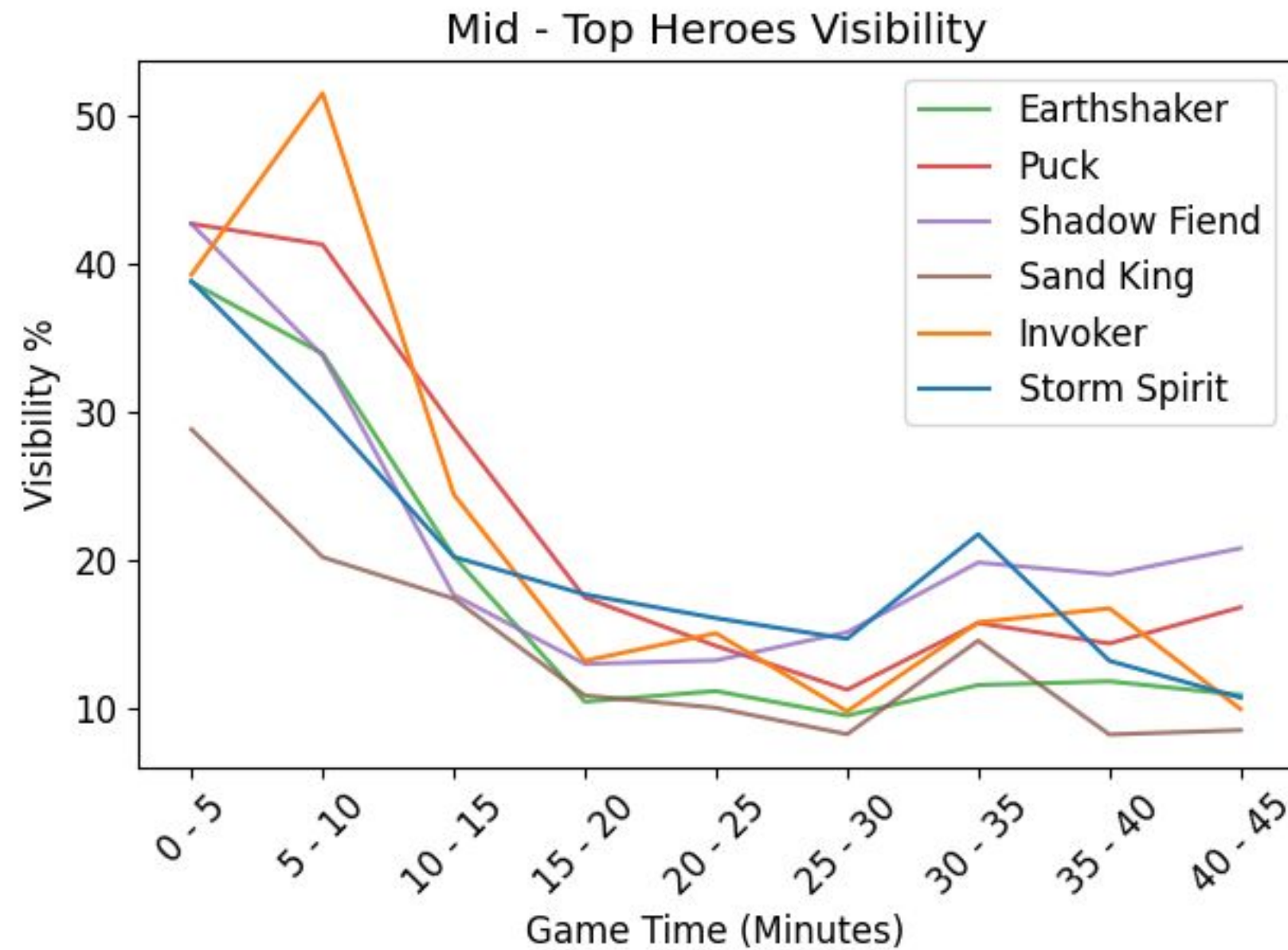
Initial Analyses

Most Played Heroes on Mid Role

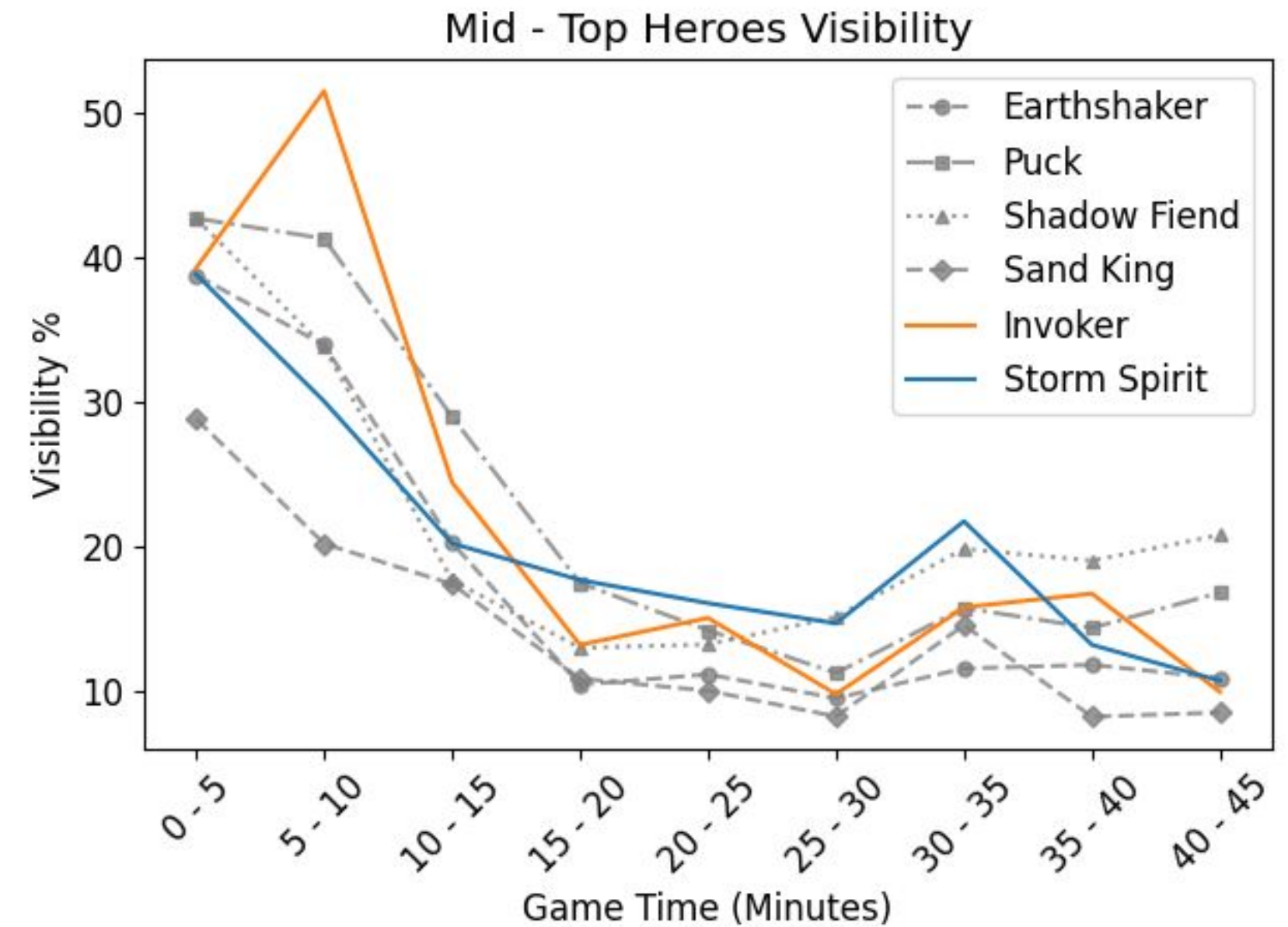


Initial Analyses

Most Played Heroes on Mid Role



Highlighting Invoker x Storm Spirit



REFERENCES

- [\[1\]](#) Tot, M., Conserva, M., Chitayat, A. P., Kokkinakis, A., Patra, S., Demediuk, S., Munoz, A. C., Olarewaju, O., Ursu, M., Kirmann, B., Hook, J., Block, F., Drachen, A., e Perez- Liebana, D. (2021). What are you looking at? Team fight prediction through player camera. In IEEE Conference on Games (CoG), pages 1–8
- [\[2\]](#) Costa, L. M., Drachen, A., Souza, F. C. M., e Xexéo, G. (2024). Artificial intelligence in moba games: A multivocal literature mapping. IEEE Transactions on Games, 16(2):250–269
- [\[3\]](#) Kim, D.-W., Park, S.-Y., Yang, S.-I., e Lee, S.-K. (2025). Real-time player tracking framework on moba game video through object detection. IEEE Transactions on Games, 17(2):498–509.





THANK YOU!

ricardo.rcarvalho@ppgia.edu.br



PUCPR
GRUPO MARISTA

